FeatureBot: Feature Engineering with AI Guidance

Industrial Capstone Project

[Project Banner: AI-Guided Feature Engineering for Adult Income Classification]

Department: Business Intelligence & Analytics
Project Code: FB-FE-2025-Q3
Priority: High
Timeline: 4 weeks
Client: [Client/Org Name]
Prepared by: [Your Name]
Date: [Insert Date]
Version: 1.0

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PROJECT OVERVIEW

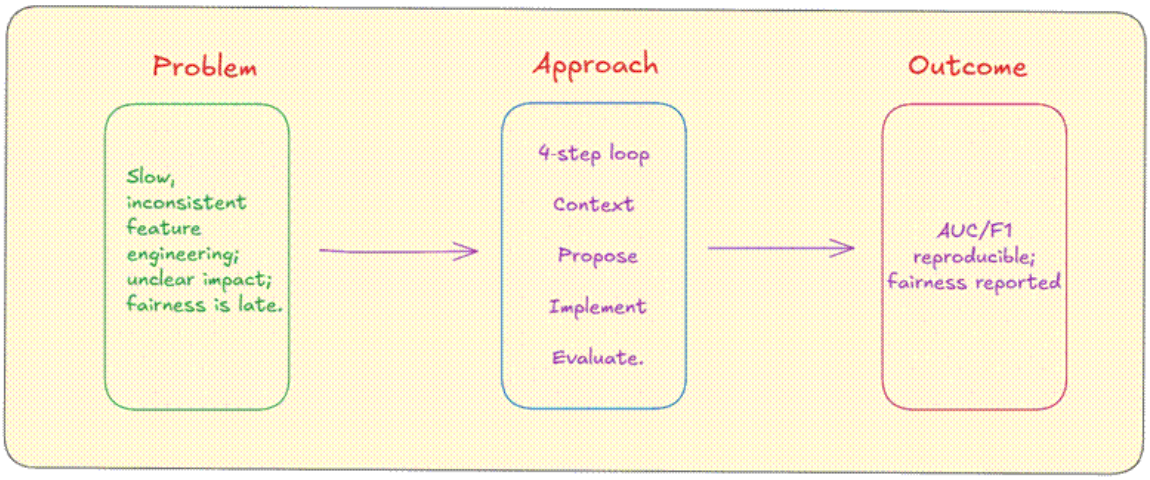
- Purpose: Build an AI-guided feature engineering workflow (FeatureBot) that accelerates discovery, evaluation, and selection of predictive features for income classification (<=50K vs >50K) using the UCI Adult dataset.
- Outcomes:
    - Measurable performance lift over a strong baseline.
    - Transparent, reproducible feature development process.
    - Fairness diagnostics and documented mitigation decisions.
- Scope: Google Colab workflow + ChatGPT prompting, feature registry, baseline and enhanced models, evaluation on holdout test, client-ready documentation.

CLIENT BACKGROUND

- Industry: Data Science
- Team Profile: BI & Analytics stakeholders seeking faster iteration on tabular ML problems and repeatable feature engineering approaches.
- Constraints: Preference for Colab-based development; lightweight tooling; document deliverables in .docx format.

BUSINESS PROBLEM STATEMENT

- The client needs to improve predictive performance and reduce development time for structured data models. Traditional manual feature engineering is slow, inconsistent, and hard to audit for fairness. A guided, repeatable process is required to generate and validate features efficiently while ensuring responsible AI practices.



USER STORY

As a Data Scientist responsible for model performance, I want an AI-guided feature engineering workflow with actionable suggestions so that I can efficiently create high-value features, improve accuracy, and maintain fairness and reproducibility.

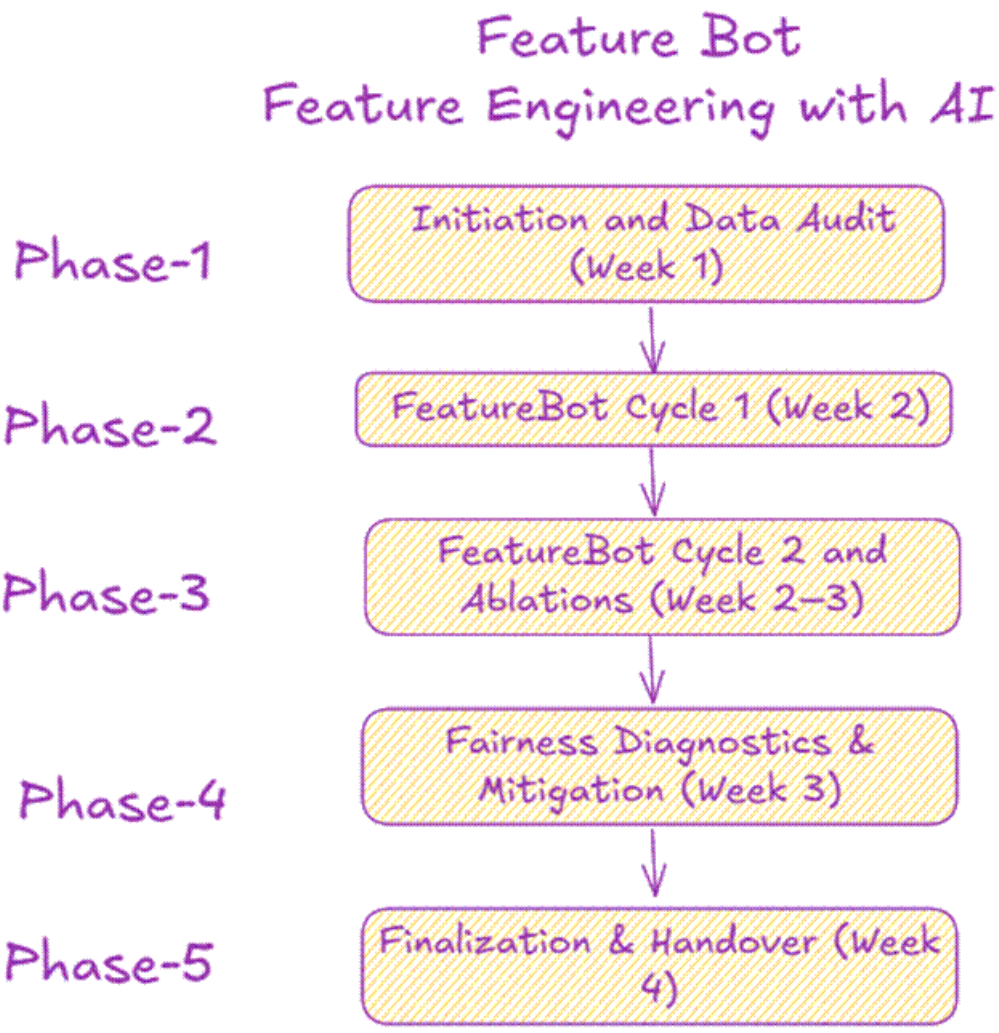
ACCEPTANCE CRITERIA

- FeatureBot loop runs at least two full cycles, each producing ≥5 actionable feature proposals with rationale and pandas pseudocode.
- Enhanced model achieves +3–5 points improvement in AUC or F1 over baseline on validation and remains stable on holdout.
- A feature registry is delivered with name, definition, type, dependencies, leakage risk, and performance deltas.
- Fairness report includes subgroup metrics (e.g., sex, race) and at least one mitigation attempt with documented trade-offs.
- Colab notebook is reproducible (fixed seeds, deterministic splits, no leakage) and includes clear instructions.
- All specified deliverables are provided in .docx/notebook/CSV formats.

PROJECT OBJECTIVES

- Implement FeatureBot: iterative propose → implement → evaluate → select loop guided by structured prompting.
- Establish governance: feature registry and experiment log with metric deltas and notes.
- Ensure reproducibility: deterministic splits, train-only fitting of preprocessors, documented prompts and responses.
- Provide fairness diagnostics and document mitigation strategies.

PROJECT TASKS (PHASE-WISE, BASED ON DATASET)



Phase 1 — Initiation and Data Audit (Week 1)

- Setup Colab environment and repository structure.
- Load adult.csv; normalize column names and target labels; handle “?” as missing.
- EDA: value counts, missingness matrix, class balance, numeric distributions/skew.
- Baseline pipeline: Logistic Regression with OHE + scaling; initial validation metrics.
- Artifacts: Initial EDA summary; baseline metrics; data quality checklist.

Phase 2 — FeatureBot Cycle 1 (Week 2)

- Prepare prompt Template A with schema, EDA highlights, and baseline errors.

- Generate ≥5 feature proposals (transformations, bins, interactions, encodings) with leakage checks.
- Implement selected features in add_feature_block with toggles.
- Evaluate via stratified K-fold; log deltas in experiment log and registry.
- Prune low-impact features based on stability and effect size.

Phase 3 — FeatureBot Cycle 2 and Ablations (Week 2–3)

- Prepare prompt Template B focused on reducing false negatives.
- Add targeted interactions, indicators, or groupings; run ablations (with/without subsets).
- Validate robustness: monitor variance across folds and holdout consistency.
- Update registry; document rejected features and reasons (e.g., instability, fairness risk).

Phase 4 — Fairness Diagnostics & Mitigation (Week 3)

- Compute subgroup metrics (TPR/FPR/Precision/Recall/AUC) by sex, race, marital-status.
- Apply one mitigation (e.g., exclude sensitive features, adjust thresholds per group).
- Document performance–fairness trade-offs and final inclusion choices.

Phase 5 — Finalization & Handover (Week 4)

- Train final model on train+validation; evaluate on holdout test.
- Produce final reports: performance summary, registry, fairness report, and lessons learned.
- Deliver .docx project document, Colab notebook, CSV registry, and experiment logs.

DELIVERABLES

- Project Document (.docx): This document with all sections and acceptance criteria.
- Google Colab Notebook: End-to-end pipeline and FeatureBot cycles with prompts and results.

- Feature Registry (CSV): name, definition, type, dependencies, leakage_risk, added_in_cycle, delta_auc, delta_f1, notes.
- Experiment Log (CSV): config_id, features_enabled summary, CV metrics, holdout metrics, runtime, comments.
- Final Evaluation Summary: Baseline vs enhanced metrics, confusion matrices, ROC/PR summaries.
- Fairness Report: Subgroup metrics and at least one mitigation attempt.

TECHNICAL REQUIREMENTS

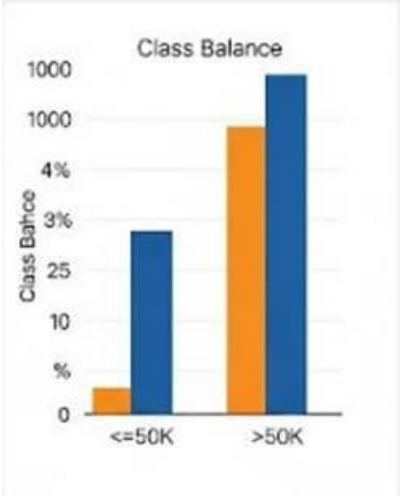
- Environment: Google Colab, Python 3.x.
- Libraries: pandas, numpy, scikit-learn, category_encoders, shap (optional).
- Reproducibility: Fixed seeds, stratified splits (60/20/20), train-only fitting of imputers/encoders, no data leakage.
- Runtime: Fits within Colab free-tier constraints; batch experiments and compact models.

DATASET INFORMATION

- Dataset: UCI Adult Income (Kaggle mirror).
- Target: Income (<=50K vs >50K).

- Attributes: age, workclass, fnlwgt, education, education-num, marital-status, occupation, relationship, race, sex, capital-gain, capital-loss, hours-per-week, native-country.
- Known considerations:
    - Missing values encoded as “?” in workclass, occupation, native-country.
    - Class imbalance (~24% >50K typical).
    - Skewed numeric variables (capital-gain, capital-loss, fnlwgt).
    - Sensitive attributes: sex, race, marital-status.

age	workclass	fnlwgt	education	educational-num	marital-status	occupation	relationship	race	gender	capital-gain	capital-loss	hours-per-week	native-country	income
25	Private	226802	11th	7	Never-married	Machine-op-ir	Own-child	Black	Male	0	0	40	United-States	<=50K
38	Private	89814	HS-grad	9	Married-civ-spou	Farming-fishir	Husband	White	Male	0	0	50	United-States	<=50K
28	Local-gov	336951	Assoc-acdm	12	Married-civ-spou	Protective-ser	Husband	White	Male	0	0	40	United-States	>50K
44	Private	160323	Some-college	10	Married-civ-spou	Machine-op-ir	Husband	Black	Male	7688	0	40	United-States	>50K
18	?	103497	Some-college	10	Never-married	?	Own-child	White	Female	0	0	30	United-States	<=50K
34	Private	198693	10th	6	Never-married	Other-service	Not-in-family	White	Male	0	0	30	United-States	<=50K
29	?	227026	HS-grad	9	Never-married	?	Unmarried	Black	Male	0	0	40	United-States	<=50K
63	Self-emp-not	104626	Prof-school	15	Married-civ-spou	Prof-specialty	Husband	White	Male	3103	0	32	United-States	>50K
24	Private	369667	Some-college	10	Never-married	Other-service	Unmarried	White	Female	0	0	40	United-States	<=50K
55	Private	104996	7th-8th	4	Married-civ-spou	Craft-repair	Husband	White	Male	0	0	10	United-States	<=50K
65	Private	184454	HS-grad	9	Married-civ-spou	Machine-op-ir	Husband	White	Male	6418	0	40	United-States	>50K
36	Federal-gov	212465	Bachelors	13	Married-civ-spou	Adm-clerical	Husband	White	Male	0	0	40	United-States	<=50K
26	Private	82091	HS-grad	9	Never-married	Adm-clerical	Not-in-family	White	Female	0	0	39	United-States	<=50K
58	?	299831	HS-grad	9	Married-civ-spou	?	Husband	White	Male	0	0	35	United-States	<=50K
48	Private	279724	HS-grad	9	Married-civ-spou	Machine-op-ir	Husband	White	Male	3103	0	48	United-States	>50K



EVALUATION CRITERIA

- Data preprocessing quality and approach (20%)
- Feature engineering creativity, rigor, and documentation (25%)
- Prompting strategy quality and reproducibility (15%)

- Modeling and evaluation methodology (15%)
- Fairness analysis and reporting (10%)
- Technical documentation clarity (10%)
- Project Management (5%)

GETTING STARTED

- Files:
    - adult.csv in the Colab working directory.
    - featurebot_adult_income.py as a separate code file (previously provided).
- Steps:
    0. Open Colab and upload adult.csv.
    1. Create featurebot_adult_income.py; paste the provided script; save.
    2. Run %run featurebot_adult_income.py to execute baseline and enhanced flows.
    3. Use ChatGPT with Template A/B/C to propose new features; add accepted ones to add_feature_block; rerun.
    4. Export feature_registry.csv and experiments_log.csv from featurebot_outputs directory.

SUPPORT

- Contact: [Your Name], [Email], [Phone/Slack]
- Hours: [Business hours and timezone]
- SLA:
    - Routine questions: 1–2 business days
    - Bug fixes: 2–3 business days
    - Enhancements: per agreed sprint plan
- Handover: Final walkthrough session and Q&A; follow-up support for 2 weeks post-delivery.

SOLUTION OVERVIEW

- FeatureBot combines structured prompting (ChatGPT) with automated experimentation (Colab) to create and validate high-impact features while maintaining reproducibility and fairness.

TECHNICAL APPROACH

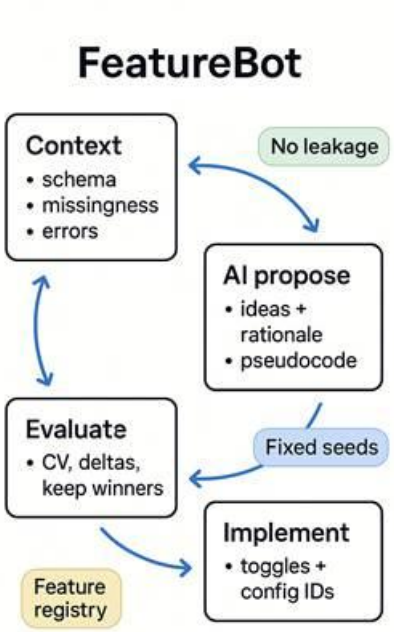
- Data Quality: Replace “?” with NaN; impute categorical (most frequent/Unknown), numeric (median); normalize names.
- Splits: Stratified 60/20/20; fixed seeds; pipelines fitted on train only.
- Pipelines: sklearn Pipeline + ColumnTransformer; leakage-safe.
- Tracking: DataFrame experiment log + CSV feature registry with deltas vs baseline.

FEATURE ENGINEERING STRATEGY

Feature	Why it helps
`capital_gain_gt0` `gain > 0`	Handles sparsity; flags non-zero gains.
`age_bin`	Captures nonlinearity for linear models.
`hours_bin`	Overtime signal; stronger separation
`age × education_num`	Seniority/skill proxy

- Encodings:
    - One-hot for low-cardinality (sex, race, relationship groups).
    - Target/CatBoost/Frequency for medium/high (occupation, native-country).
- Numeric Transforms:
    - log1p for capital-gain/loss, fnlwgt.
    - Binning for age and hours-per-week.
- Indicators:
    - capital_gain_gt0, capital_loss_gt0, overtime (hours > 40).
- Interactions:
    - age × education-num; hours-per-week × occupation; marital-status × relationship; gain>0 × occupation; workclass × hours-bin; sex × occupation (audited for fairness).
- Groupings:
    - Education buckets (HS-or-less, Some-College, Bachelors, Advanced).
    - Region buckets for native-country (Americas, Europe, Asia, Other).
- Selection:
    - Univariate filters, model-based importances, stepwise addition with CV stability.

AI-GUIDED WORKFLOW (FEATUREBOT)



- Cycle

    1. Provide schema, value counts, missingness, baseline metrics, top errors.
    2. Receive prioritized feature proposals with rationale, leakage flags, and pandas pseudocode.
    3. Implement features via toggles; log configuration ID.
    4. Evaluate with stratified K-fold; record deltas; prune low-impact ideas.

- Prompt Templates
    - A: Schema + EDA → 5 features with rationale, leakage risk, pandas pseudocode; optimize for F1 on >50K.
    - B: False negatives → 3 interactions to reduce FN; provide code and fairness notes.
    - C: Subgroup metrics → mitigations minimizing overall F1 loss; include steps.
- Governance
    - Feature Registry fields: name, definition, type, dependencies, leakage risk, added_in_cycle, delta AUC/F1, notes.
- Why it's useful:
    - Faster idea generation
    - Clear, measurable improvements (keep only winners)
    - Reproducible and documented
    - Fairness stays visible with subgroup metrics

One-line summary:

FeatureBot = ChatGPT for smart feature ideas + Colab for testing them, so you quickly find and keep the features that actually boost your model.

MODELING & EVALUATION

- Baselines: Logistic Regression (L2), RandomForest, Gradient Boosting/XGBoost/LightGBM (if available).
- Metrics: Primary—ROC AUC, F1 (>50K). Secondary—Precision/Recall, PR AUC, calibration (Brier), confusion matrix; subgroup metrics.
- Validation: Stratified 5-fold; threshold chosen on validation; final report on holdout.
- Interpretability: Global importances; permutation importance; optional SHAP.

BIAS & FAIRNESS

Subgroup	F1	AUC
Sex: Male	0.85	0.92
Sex: Female	0.83	0.91

Document any mitigation and trade-offs.

- Sensitive attributes: sex, race, marital-status.
- Diagnostics: TPR/FPR/Precision/Recall/AUC by subgroup; parity ratios.
- Mitigations: Exclude sensitive vars; constrain influence or apply group thresholds; document trade-offs.

TECHNICAL DOCUMENTATION FOR ANALYTICS TEAM

- Environment: Colab; Python 3.x; pandas, numpy, scikit-learn, category_encoders, shap (optional).
- Notebook Sections:
    1. Setup & data load
    2. EDA summary
    3. Baseline pipeline
    4. FeatureBot cycles (prompts + implementation)
    5. Evaluation, ablations, fairness
    6. Final model & holdout report
- Experiment Tracking: config_id, features_enabled, model, CV metrics, holdout metrics, runtime, comments.

- QA: Train-only fitting; leakage checks; fixed seeds; holdout integrity.

DATA REQUIREMENTS

- Files: adult.csv (or equivalent).
- Columns: As listed; ensure consistent dtypes; normalize target labels (>50K, <=50K, including dotted variants).
- Volume: ~48K rows; suitable for Colab memory limits.
- Permissions: Public dataset; no PII beyond provided attributes.

QUALITY & ACCEPTANCE CRITERIA

- Preprocessing robust; no data leakage.
- FeatureBot ≥2 cycles; registry populated with impactful features and deltas.
- Performance improvement +3–5 AUC or F1 on validation; consistent holdout performance.
- Fairness: subgroup metrics with at least one mitigation attempt.
- Documentation: reproducible notebook; exportable registry; final summary.

PROJECT PLAN & TIMELINE

- Week 1: Data audit, EDA, baseline models, initial prompts.
- Week 2: FeatureBot cycles 2–3; interim report; registry v1.
- Week 3: Fairness diagnostics; ablations; finalize feature set.
- Week 4: Final model; documentation; handover.

RISKS & MITIGATIONS

- Data quality and skew: Standardized cleaning; log transforms; QA checks.
- Overfitting: K-fold; holdout; small-batch feature additions.
- Fairness risks: Subgroup reporting; exclusion/constraints; threshold tuning.
- Colab limits: Compact models; batching; caching encodings.

- Prompt variability: Fixed templates; rich EDA context; human review and pruning.

COLAB STARTER CODE

Use the separate code file featurebot_adult_income.py I provided earlier. In Colab:

- Upload adult.csv
- Create the file featurebot_adult_income.py and paste the script
- Run: %run featurebot_adult_income.py
- Iterate: Add new features into add_feature_block, update the registry entries, and rerun.

APPENDICES

- Prompt Template A (Features): “Given schema + EDA [paste], propose 5 transformations with rationale, leakage risk, and pandas pseudocode; optimize for F1 on >50K.”
- Prompt Template B (FN Interactions): “Given validation errors [paste], suggest 3 interactions to reduce false negatives; provide code and fairness notes.”
- Prompt Template C (Fairness): “Given subgroup metrics [paste], propose mitigations with minimal overall F1 loss; include steps.”
- Feature Registry Fields: name, definition, type, dependencies, leakage_risk, added_in_cycle, delta_auc, delta_f1, notes.